

*Source databases
(pseudonymised, 1:1 mirror)*

Cohort A — community adults
7 waves · 122,575 records
30,677 adults

Cohort B — health management
4 waves · 21.7 M lab rows
71,209 person-years

Cohort C — regional examination
3 waves · 1,389,967 records
54,490 death records

Detect–Quantify–Repair–Verify closed loop

1 · DETECT
15 physiological signatures
+ cross-year median scan
+ correlation probes

2 · QUANTIFY
before/after contrasts
screening counts, medians
fabricated-diagnosis counts

3 · REPAIR
true-value remapping
copy-shift (never in-place)
nullify · deduplicate-flag

4 · VERIFY
residual signatures = 0
correlation recovery
regression · spot checks

Injection harness — 2,400 synthetic shifts
recall 0.985 · specificity 1.000
blind re-detection 0.987

Data records (this Descriptor)

Cleaned analysis layers
DR1–DR3 · controlled access
person-wave / person-year
tables + screening vars

Governance artefacts
DR4 · open
signature library · audits
repair reports · harness

technical validation (Fig. 3)